



ĐẠI HỌC ĐÀ NẴNG

TRƯỜNG ĐẠI HỌC BÁCH KHOA

D
BACH KHOA

TRÍ TUỆ NHÂN TẠO

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G



Khoa Công Nghệ Thông Tin

TS. Nguyễn Văn Hiệu

TRÍ TUỆ NHÂN TẠO

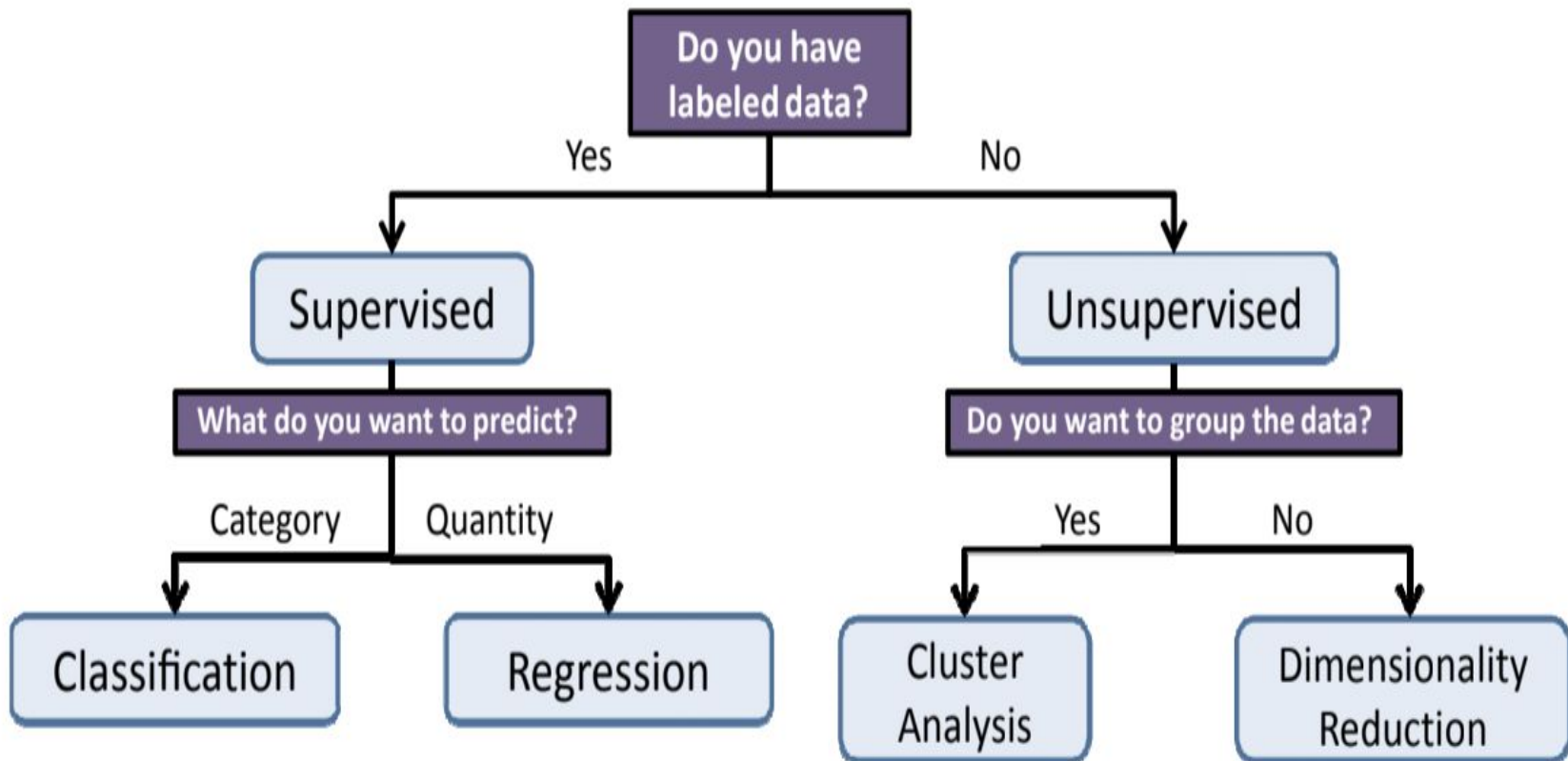
Chương 6: Perceptron

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Nội dung

- Giới thiệu
- Perceptron
- Demo



Phân lớp

Given: Training data: $(x_1, y_1), \dots, (x_n, y_n) / x_i \in \mathbb{R}^d$ and y_i is discrete (categorical/qualitative), $y_i \in \mathbb{Y}$.

Example $\mathbb{Y} = \{-1, +1\}$, $\mathbb{Y} = \{0, 1\}$.

Task: Learn a classification function:

$$f : \mathbb{R}^d \longrightarrow \mathbb{Y}$$

Linear Classification: A classification model is said to be linear if it is represented by a linear function f (linear hyperplane)

Phân lớp: Ví dụ

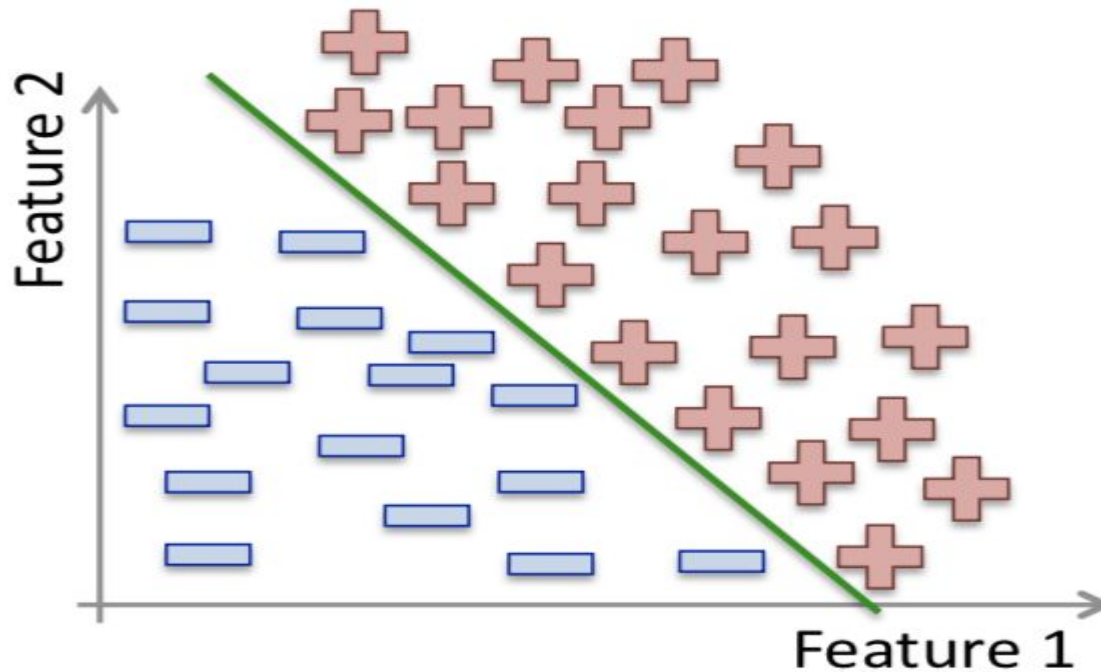
1. Email Spam/Ham → Which email is junk?
2. Tumor benign/malignant → Which patient has cancer?
3. Credit default/not default → Which customers will default on their credit card debt?

Balance	Income	Default
300	\$20,000.00	no
2000	\$60,000.00	no
5000	\$45,000.00	yes
.	.	.
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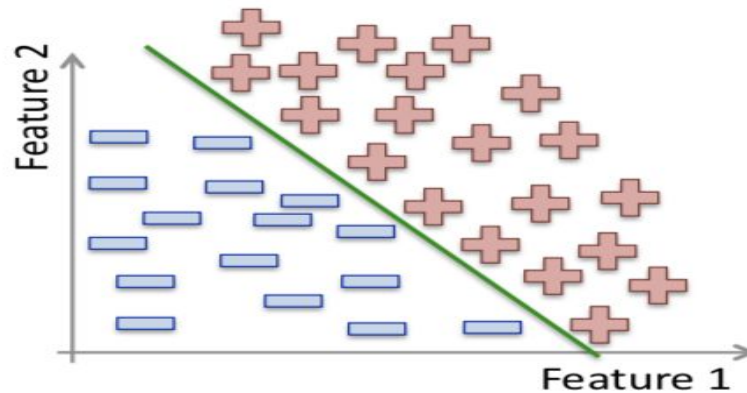
Perceptron

- Belongs to Neural Networks class of algorithms (algorithms that try to mimic how the brain functions).
- The first algorithm used was the Perceptron (Rosenblatt 1959).
- Worked extremely well to recognize:
 1. handwritten characters (LeCun et al. 1989),
 2. faces (Cottrell 1990)
- NN were popular in the 90's but then lost some of its popularity.
- Now NN back with deep learning.

Perfectly separable data

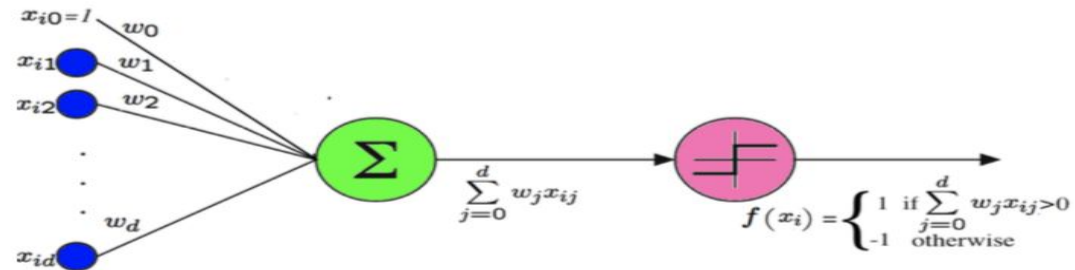


Perceptron



- Linear classification method.
- Simplest classification method.
- Simplest neural network.
- For perfectly separated data.

Perceptron



Given n examples and d features.

$$f(x_i) = \text{sign}\left(\sum_{j=0}^d w_j x_{ij}\right)$$

Perceptron

$$label(x) = \begin{cases} +1 & w^T x > 0, \\ -1 & \end{cases}$$

$$label(x) = \text{sign}(w^T x)$$

$$\text{Loss function : } J(w) = \sum_{x_i \in M} (-y_i \text{sign}(w^T x_i))$$

$$\text{New loss function : } J(w) = \sum_{x_i \in M} (-y_i w^T x_i)$$

$$J(w, x_i, y_i) = -y_i w^T x_i$$
$$\frac{dJ(w, x_i, y_i)}{dw} = -y_i x_i$$

Perceptron

- Works perfectly if data is linearly separable. If not, it will not converge.
- Idea: Start with a random hyperplane and adjust it using your training data.
- Iterative method.

Perceptron

Perceptron Algorithm

Input: A set of examples, $(x_1, y_1), \dots, (x_n, y_n)$

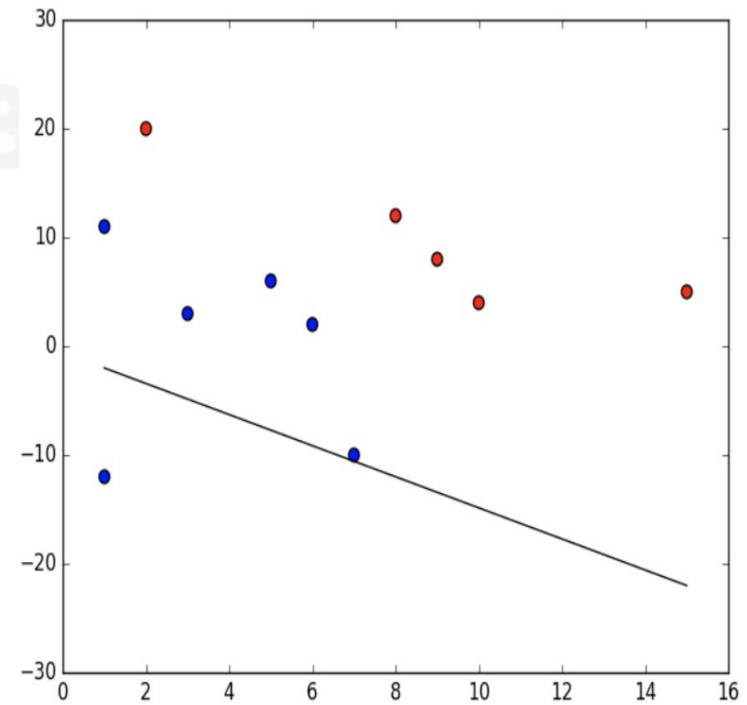
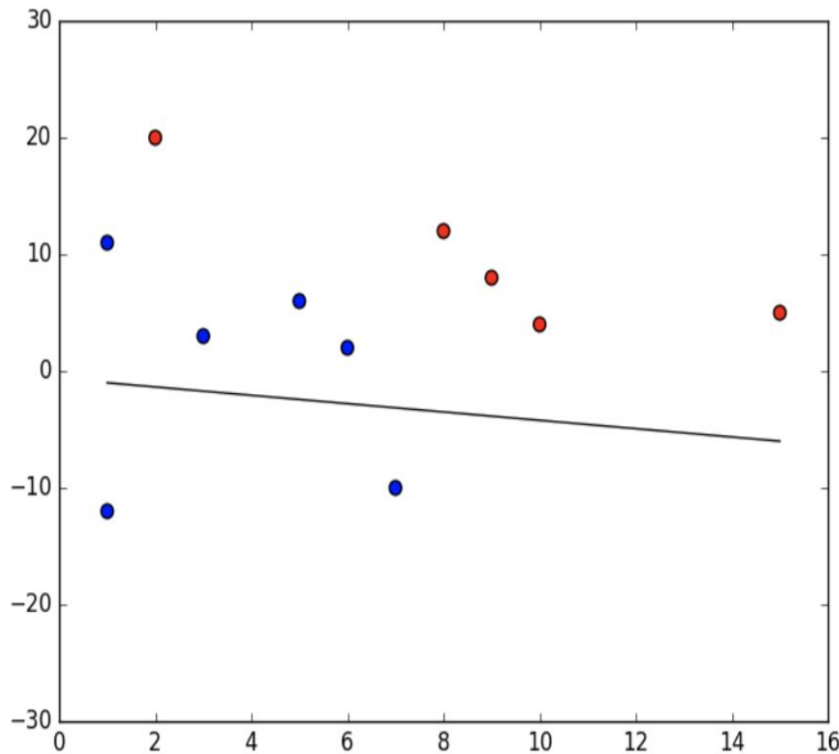
Output: A perceptron defined by $(w_0, w_1, \dots, w_d)$

Begin

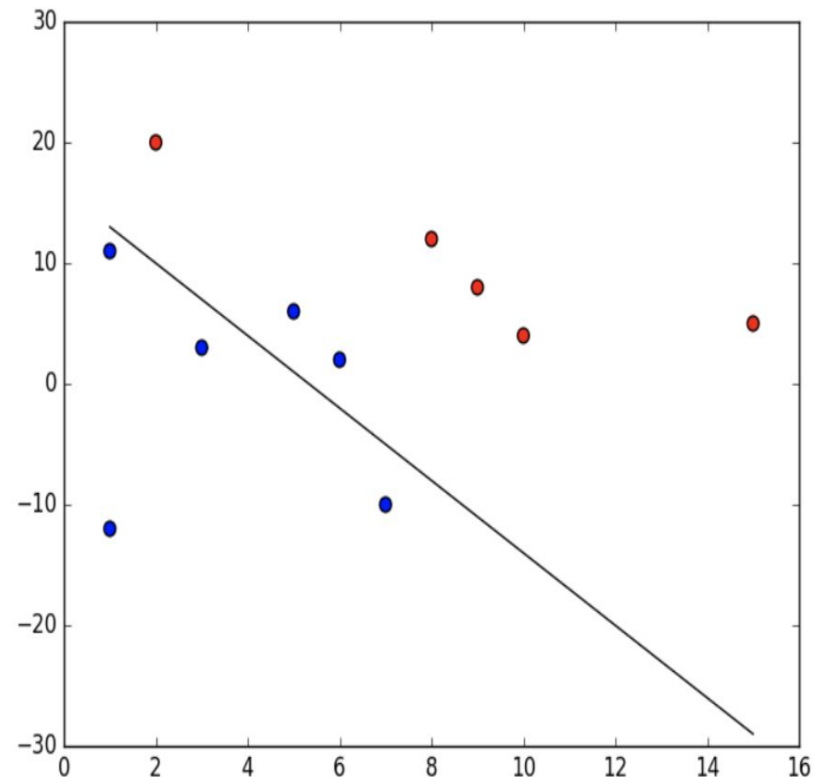
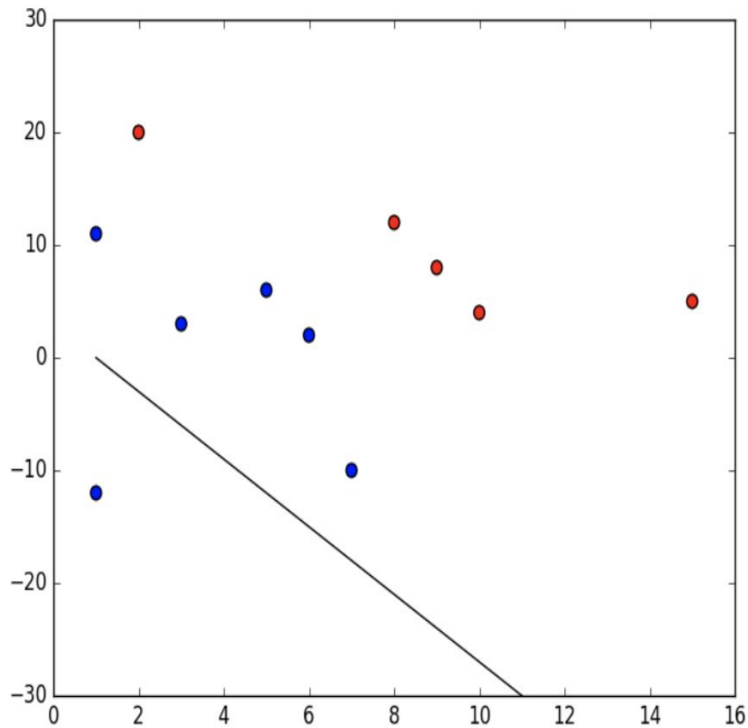
2. Initialize the weights w_j to 0 $\forall j \in \{0, \dots, d\}$
3. Repeat until convergence
4. For each example $x_i \forall i \in \{1, \dots, n\}$
5. if $y_i f(x_i) \leq 0$ #an error?
6. update all w_j with $w_j := w_j + y_i x_i$ #adjust the weights

End

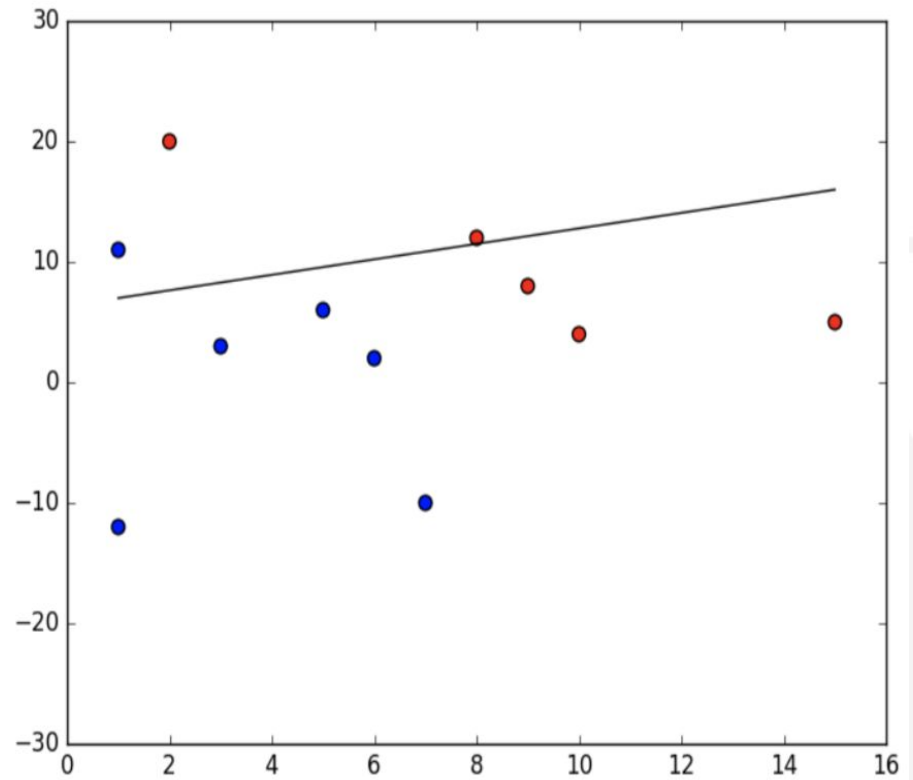
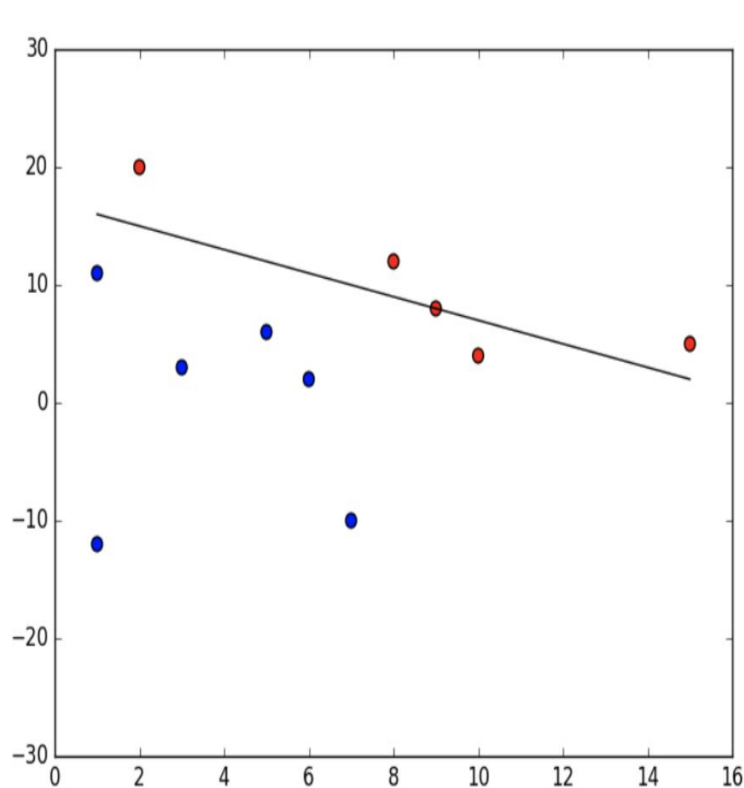
Perceptron: Example



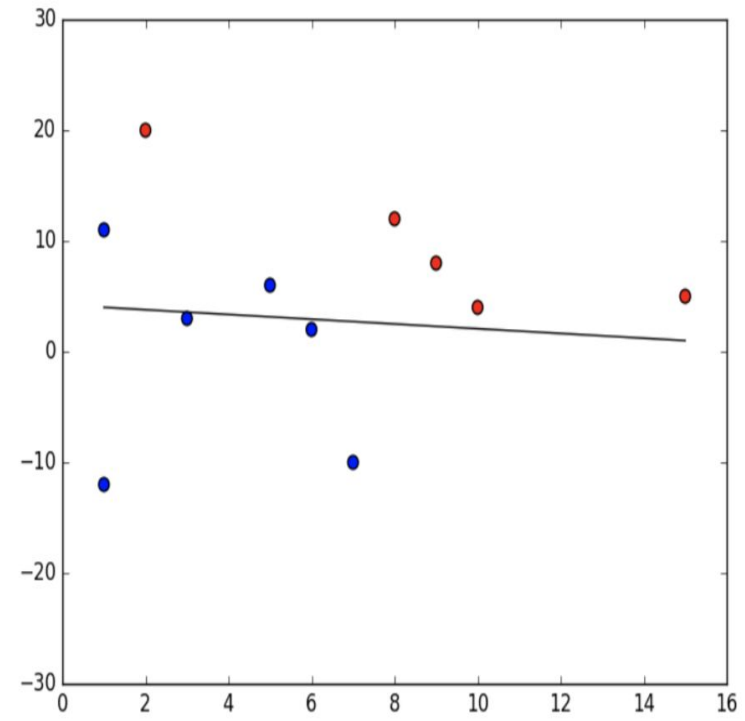
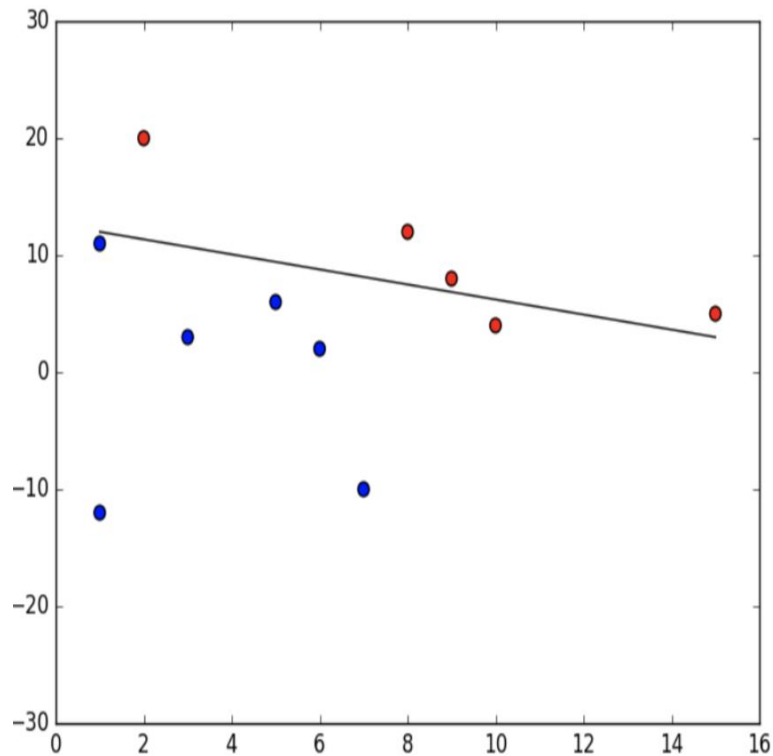
Perceptron: Example



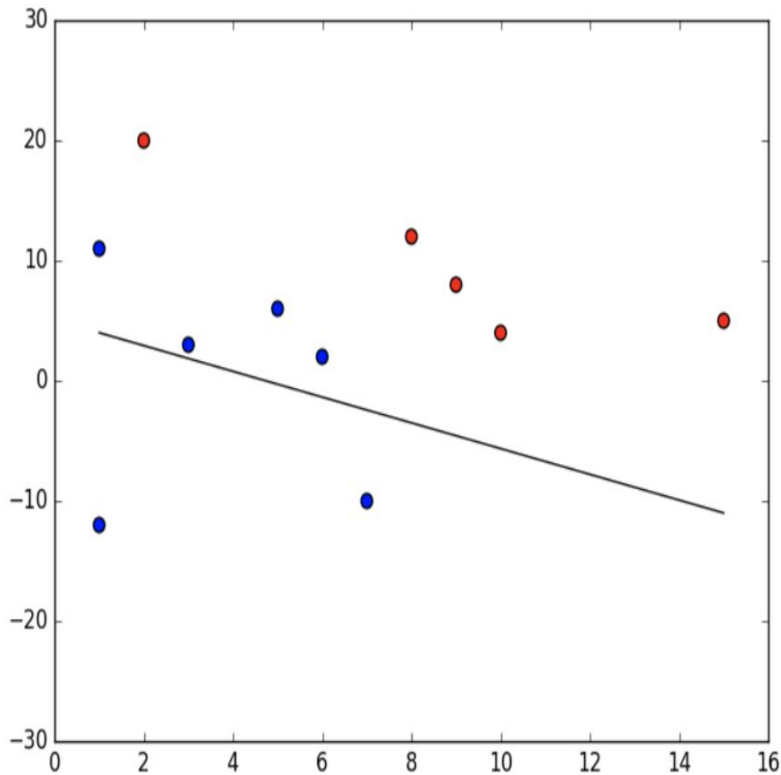
Perceptron: Example



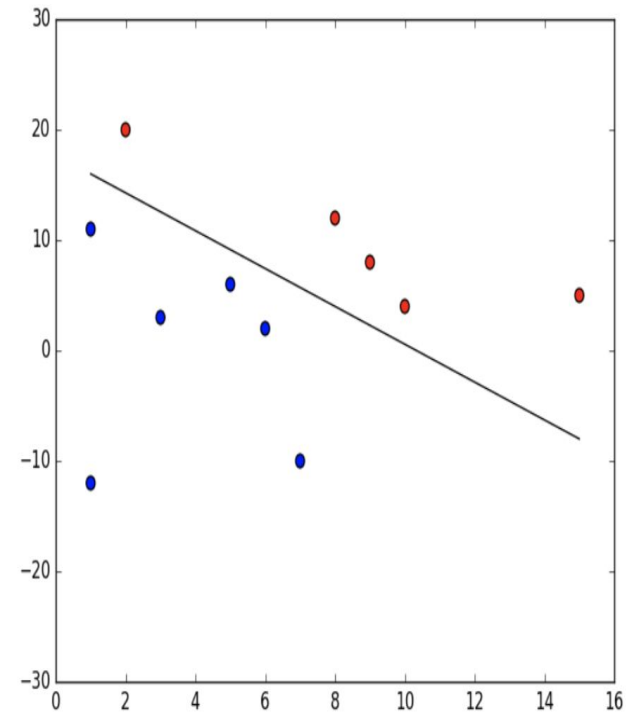
Perceptron: Example



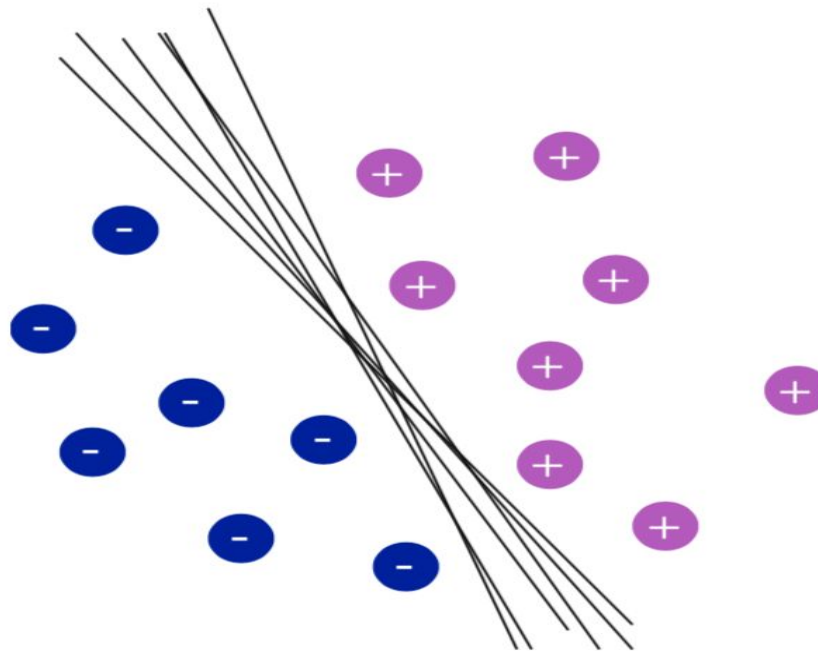
Perceptron: Example



Finally converged!



Choice of the hyperplane



Lots of possible solutions!

Digression: Idea of SVM is to find the optimal solution.



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